

MIAE 221: Materials Science for Engineers

Part 1: Materials Classes, Atomic Architecture & Interatomic Energy Curves

1. The Spectrum of Engineering Materials

Every object engineered by humankind—from microchips and jet turbine blades to biomedical stents and composite golf shafts—derives its capabilities directly from the arrangement of its atoms and the bonds holding them together. In engineering, materials are classified into three primary categories, along with advanced composites and semiconductors:

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graph TD
    EM["ENGINEERING MATERIALS"] --> M["METALS<br/>(Fe, Al, Cu, Ti)"]
    EM --> C["CERAMICS<br/>(Al2O3, SiC, SiO2)"]
    EM --> P["POLYMERS<br/>(PE, PTFE, Nylon)"]
    M --> COMP["COMPOSITES<br/>(CFRP, Fiberglass, MMC)"]
    C --> COMP
    P --> COMP
```

Comparative Engineering Property Matrix

Property	Metals	Ceramics	Polymers	Composites
Bonding Type	Metallic (sea of electrons)	Ionic & Covalent	Covalent (chains) + Secondary	Combined (Matrix + Fiber)
Ductility (Deformability)	High (readily bend/stretch)	Very Low (extremely brittle)	High to Variable	Low to Moderate
Elastic Modulus (Stiffness)	High (100–400 GPa)	Very High (150–500 GPa)	Low (0.1–5 GPa)	Tailorable (50–250 GPa)
Yield / Tensile Strength	Medium to High	High compressive, poor tensile	Low to Medium	Very High along fiber axis
Fracture Toughness (K_{Ic})	High (resists crack growth)	Poor (catastrophic fracture)	Moderate to High	High (fiber pullout)
Thermal & Electrical Cond.	High (free electrons)	Very Low (insulators)	Very Low (insulators)	Tailorable
Temperature Resistance	Moderate to High	Exceptional (refractories)	Poor (melts/degrades early)	Moderate

Why Do Materials Behave Differently? An Intuitive Analogy

1. **Metals (The Ballroom Analogy):** Imagine a crowded ballroom where everyone lets go of their personal coats and tosses them into the center. The coats (valence electrons) form a mobile, flowing cloud, while the people (positively charged ion cores) move past one another effortlessly without destroying the party. When you bend a metal wire, the atomic planes slide across each other without shattering because the electron "sea" immediately adapts and shields the positively charged cores. This gives metals **high ductility** and **electrical conductivity**.
2. **Ceramics (The Magnetic Grid Analogy):** Imagine a tight checkerboard of alternating positive and negative bar magnets glued together in rigid geometric rows. The bonds are ferocious, giving ceramics **extraordinary hardness** and **high melting points**. However, if you attempt to slide one layer by just one atomic spacing, like charges are forced directly adjacent to like charges ($+$ $\leftrightarrow$ $+$ and $-$ $\leftrightarrow$ $-$). The repulsive electrostatic shock violently shatters the material along crystal cleavage planes. This is why ceramics are **unforgivingly brittle**.
3. **Polymers (The Cooked Spaghetti Analogy):** Polymers consist of gigantic hydrocarbon chains (macromolecules) where carbon atoms form exceptionally strong covalent backbones. Between adjacent spaghetti noodles, however, only weak secondary bonds (van der Waals or hydrogen bonds) exist. When you pull on plastic, the noodles slide and untangle, giving polymers **low stiffness, low melting temperatures,** and **high flexibility**.

2. Atomic Architecture & The Quantum Mechanical Model

To predict how atoms interact, we must examine how electrons arrange themselves around the nucleus.

Model Characteristic	Bohr Model (1913)	Modern Quantum Mechanical Model
Electron Motion	Fixed 2D planar circular orbits ($n = 1, 2, 3 \dots$)	3D spatial wavefunctions / orbitals (Ψ)
Position Certainty	Precise radius r and deterministic velocity	Heisenberg Uncertainty ($\Delta x \cdot \Delta p \geq \hbar/2$); probability density ρ
Spatial Shapes	Strictly circular or elliptical rings	Complex geometric probability lobes (s : sphere, p : dumbbell, d : cloverleaf)
Governing Mathematics	Quantized angular momentum $L = n\hbar$	Schrödinger Wave Equation $\hat{H}\Psi = E\Psi$

The Evolution of Atomic Understanding

- **The Bohr Model (1913):** Postulated that electrons circle the nucleus in fixed, quantized circular tracks characterized by discrete quantum energy levels ($n = 1, 2, 3 \dots$). While useful for explaining hydrogen emission lines, it failed to account for multi-electron atoms or orbital shapes.
- **The Wave-Mechanical Model (Modern Quantum Mechanics):** According to the de Broglie hypothesis and Heisenberg Uncertainty Principle, electrons behave simultaneously as particles and wavefunctions (Ψ). We

cannot specify an exact trajectory; instead, electrons inhabit three-dimensional probability density distributions known as **orbitals**, where the probability of finding an electron in a given volume is $P = |\Psi|^2$.

The Four Quantum Numbers

Every electron in an atom is uniquely identified by four quantum numbers:

1. **Principal Quantum Number (n):**

- Takes integer values: $n = 1, 2, 3, 4, \dots$
- Dictates the **major electron shell** and represents average radial distance from the nucleus and primary energy level.

2. **Azimuthal / Orbital Angular Momentum Quantum Number (l):**

- Takes integer values: $l = 0, 1, 2, \dots, (n - 1)$
- Dictates the **subshell geometric shape**:
 - $l = 0 \implies$ **s** orbital (spherical)
 - $l = 1 \implies$ **p** orbital (dumbbell-shaped)
 - $l = 2 \implies$ **d** orbital (cloverleaf-shaped)
 - $l = 3 \implies$ **f** orbital (complex multi-lobed)

3. **Magnetic Quantum Number (m_l):**

- Takes integer values: $m_l = -l, \dots, 0, \dots, +l$
- Dictates the **spatial orientation** of the orbital in space.
- Total orientations per subshell: $2l + 1$ (e.g., *s* has 1, *p* has 3, *d* has 5, *f* has 7).

4. **Spin Quantum Number (m_s):**

- Takes two values: $m_s = +\frac{1}{2}$ ($\uparrow$, spin-up) or $m_s = -\frac{1}{2}$ ($\downarrow$, spin-down).
- Represents intrinsic angular momentum (electron spin).

Governing Rules of Electron Configurations

1. **Aufbau Principle ("Building Up"):** Electrons fill lower-energy orbitals before occupying higher-energy ones:

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \dots$$

[!NOTE] Notice that the *4s* orbital fills

before

the *3d* orbital because *4s* has slightly lower electrostatic energy in neutral atoms!

2. **Pauli Exclusion Principle:** No two electrons in an atom can have the exact same set of all four quantum numbers (n, l, m_l, m_s). Consequently, **each spatial orbital can hold a maximum of 2 electrons** with opposite spins ($\uparrow\downarrow$).

3. **Hund's Rule of Maximum Multiplicity:** Within a degenerate subshell (such as the three *p* orbitals or five *d* orbitals), electrons occupy separate empty orbitals with parallel spins before pairing up, minimizing mutual electrostatic repulsion.

3. The Periodic Table & Fundamental Trends

The layout of the modern periodic table directly reflects the filling of atomic subshells:

Group	Category	Valence Subshell	Typical Elements	Primary Chemical Character
Group IA	Alkali Metals	s^1	Li, Na, K, Rb, Cs	Highly electropositive; loses 1 electron (+1)
Group IIA	Alkaline Earth	s^2	Be, Mg, Ca, Sr, Ba	Electropositive; loses 2 electrons (+2)
Groups IIIB–IIB	Transition Metals	$(n-1)d^{1-10}, ns^2$	Fe, Ni, Cu, Ti, Cr	Variable oxidation states; metallic bonding sea
Group IIIA–VIA	Post-transition / Metalloids	s^2p^{1-4}	Al, Si, Ge, C, O	Covalent & directional bonding tendencies
Group VIIA	Halogens	s^2p^5	F, Cl, Br, I	Highly electronegative; gains 1 electron (−1)
Group 0 (VIII A)	Noble / Inert Gases	s^2p^6	He, Ne, Ar, Kr, Xe	Stable full octet; inert, zero electronegativity

Periodic Chemical Families

1. Inert Gases (Noble Gases, Group 0 / Group VIII A):

- Full outer valence shell (s^2p^6 , or $1s^2$ for He).
- Exceptionally stable electron configurations; negligible chemical reactivity; high ionization energy; electronegativity essentially zero.

2. Halogens (Group VII A):

- Configuration: s^2p^5 (one electron short of a full octet).
- Highest electron affinity; aggressively attract electrons to form stable −1 anions (F^- , Cl^- , Br^-).

3. Alkali Metals (Group IA):

- Configuration: s^1 (one loosely held valence electron outside a noble gas core).
- Lowest ionization energy; readily lose their valence electron to form +1 cations (Li^+ , Na^+ , K^+).

4. Alkaline Earth Metals (Group IIA):

- Configuration: s^2 ; readily lose 2 valence electrons to form +2 cations (Mg^{2+} , Ca^{2+} , Ba^{2+}).

5. Transition Metals (Groups IIIB through IIB):

- Incompletely filled inner d -orbitals (d^1 to d^{10}) with outer s^2 electrons.
- Exhibit multiple oxidation states and form metallic bonds through delocalized d - and s -electrons.

Core Periodic Trends

Trend	Across a Period ($\rightarrow$)	Down a Group ($\downarrow$)	Underlying Physical Mechanism
Atomic Radius	Decreases	Increases	Moving right, nuclear charge Z_{eff} pulls electrons closer. Moving down, additional principal quantum shells (n) add radial distance.
Ionization Energy	Increases	Decreases	Harder to remove electrons held by high nuclear charge; easier to remove electrons shielded by inner shells.
Electronegativity	Increases	Decreases	Measures tendency of an atom to attract electrons. Fluorine is the highest ($X_F = 4.0$); Francium/Cesium are lowest ($X_{Cs} = 0.7$).

4. Quantitative Stoichiometry & Atom Count Calculations

In engineering, we bridge microscopic atomic quantities and macroscopic physical components using the **mole** and **Avogadro's number**:

$$\text{Number of Moles } n_{mol} = \frac{m}{A} = \frac{\rho \cdot V}{A}$$

$$\text{Number of Atoms } N = n_{mol} \cdot N_A = \frac{m}{A} \cdot N_A$$

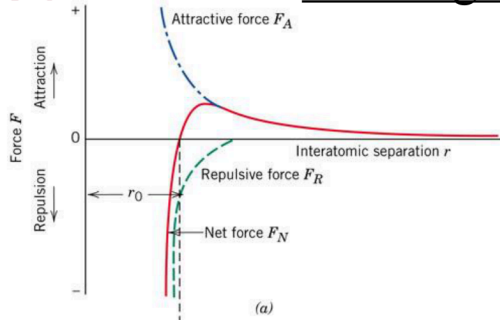
Where:

- m = Mass of the specimen (g)
- V = Geometric volume (cm^3)
- ρ = Mass density (g/cm^3)
- A = Atomic weight (g/mol)
- $N_A = 6.022 \times 10^{23}$ atoms/mol (Avogadro's constant)

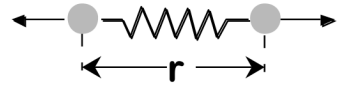
5. Interatomic Forces & Potential Energy Curves

Why do solid objects hold their shape, resist compression, and resist stretching? The answer lies in the competition between attractive and repulsive forces between adjacent atoms.

Bonding Forces and Energies

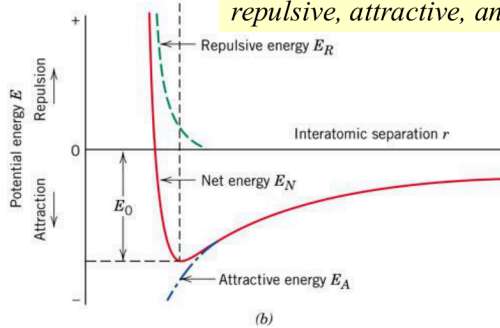


Bond length, r



Net force is given by the **sum** of an **attractive** force and a **repulsive** force

repulsive, attractive, and net forces



Potential is given by the integral of the net force curve with respect to distance:

$$E = \int F \cdot dr$$

Note: equilibrium separation occurs where the net force = 0

repulsive, attractive, and net energies

Figure 1.1: The Interatomic Potential Energy Curve $E_N(r) = E_A(r) + E_R(r)$ and net force $F_N(r) = -dE_N/dr$ (Dr. Medraj MIAE 221 Lecture 3). At equilibrium separation r_0 , net force is zero ($F_N(r_0) = 0$) and potential energy reaches minimum bonding well depth $-E_0$.

Bonding Forces and Energies

Bonding energy: Minimum of the potential vs. distance curve.

- Indicates how much energy must be supplied to completely disassociate the two atoms
- Depth of the potential well indicates bonding strength
 - Deep well → bonded
 - Shallow well → bonded

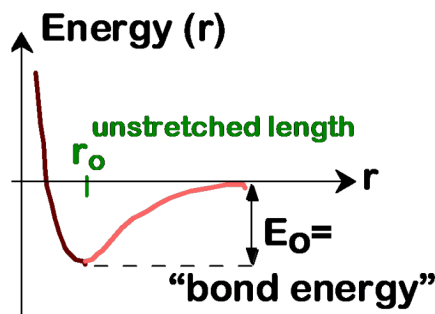


Figure 1.2: Interatomic Force Components vs. atomic separation r . Attractive Coulombic force $F_A(r) \propto -1/r^2$ dominates at large separations; quantum electron cloud overlap repulsion $F_R(r) \propto +1/r^n$ steeply resists compression at $r < r_0$.

Mathematical Formulation of Forces

When two neutral or ionized atoms approach:

1. **Attractive Force (F_A):** Arises from Coulombic electrostatic attraction between opposite charges or electron-nucleus interactions:

$$F_A(r) = -\frac{A'}{r^2} \quad \text{or generally} \quad F_A(r) = -\frac{a}{r^m}$$

2. **Repulsive Force (F_R):** Arises when electron clouds overlap, triggering quantum mechanical Pauli exclusion repulsion:

$$F_R(r) = +\frac{b}{r^p} \quad (\text{where } p > m)$$

3. **Net Force (F_N):**

$$F_N(r) = F_A(r) + F_R(r)$$

Potential Energy Integration

Potential energy is related to force by:

$$E(r) = \int_{\infty}^r F(r), dr \iff F(r) = -\frac{dE(r)}{dr}$$

Substituting the attractive and repulsive terms gives the classic **Net Interatomic Energy Equation**:

$$E_N(r) = E_A(r) + E_R(r) = -\frac{A}{r} + \frac{B}{r^n}$$

Where:

- $E_A(r) = -\frac{A}{r}$: Attractive energy (negative, stabilizing; $A > 0$).
 - $E_R(r) = +\frac{B}{r^n}$: Repulsive energy (positive, destabilizing; $B > 0, n \approx 8 - 12$).
 - r : Interatomic center-to-center separation.
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The Equilibrium State (r_0, E_0)

At equilibrium separation r_0 :

1. The attractive force exactly balances the repulsive force:

$$F_N(r_0) = 0 \iff F_A(r_0) = -F_R(r_0)$$

2. The net potential energy curve reaches its **absolute minimum**:

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

3. The depth of this minimum is the **Bonding Energy** (E_0):

$$E_0 = E_N(r_0) < 0$$

E_0 represents the exact mechanical work required to rip the two bonded atoms apart to infinite separation.

Step-by-Step Derivation of r_0 and E_0

Given the net energy function:

$$E_N(r) = -\frac{A}{r} + \frac{B}{r^n} = -Ar^{-1} + Br^{-n}$$

Step 1: Differentiate with respect to r

$$\frac{dE_N}{dr} = (-A)(-1)r^{-2} + (B)(-n)r^{-n-1} = \frac{A}{r^2} - \frac{nB}{r^{n+1}}$$

Step 2: Set derivative to zero at $r = r_0$

$$\frac{A}{r_0^2} - \frac{nB}{r_0^{n+1}} = 0 \implies \frac{A}{r_0^2} = \frac{nB}{r_0^{n+1}}$$

Step 3: Solve for equilibrium spacing r_0

Multiply both sides by r_0^{n+1} :

$$A, r_0^{n-1} = nB \implies r_0^{n-1} = \frac{nB}{A}$$
$$r_0 = \left(\frac{nB}{A} \right)^{\frac{1}{n-1}}$$

Step 4: Calculate Bonding Energy E_0

Substitute r_0 back into $E_N(r_0)$:

$$E_0 = -\frac{A}{r_0} + \frac{B}{r_0^n} = -\frac{A}{r_0} + \frac{1}{r_0} \left(\frac{B}{r_0^{n-1}} \right)$$

Since $r_0^{n-1} = \frac{nB}{A}$, we have $\frac{B}{r_0^{n-1}} = \frac{A}{n}$:

$$E_0 = -\frac{A}{r_0} + \frac{1}{r_0} \left(\frac{A}{n} \right) = -\frac{A}{r_0} \left(1 - \frac{1}{n} \right) = -\frac{A(n-1)}{n, r_0}$$

This elegant relationship shows that the bond energy E_0 is always directly proportional to the attractive constant A and inversely proportional to the equilibrium spacing r_0 .